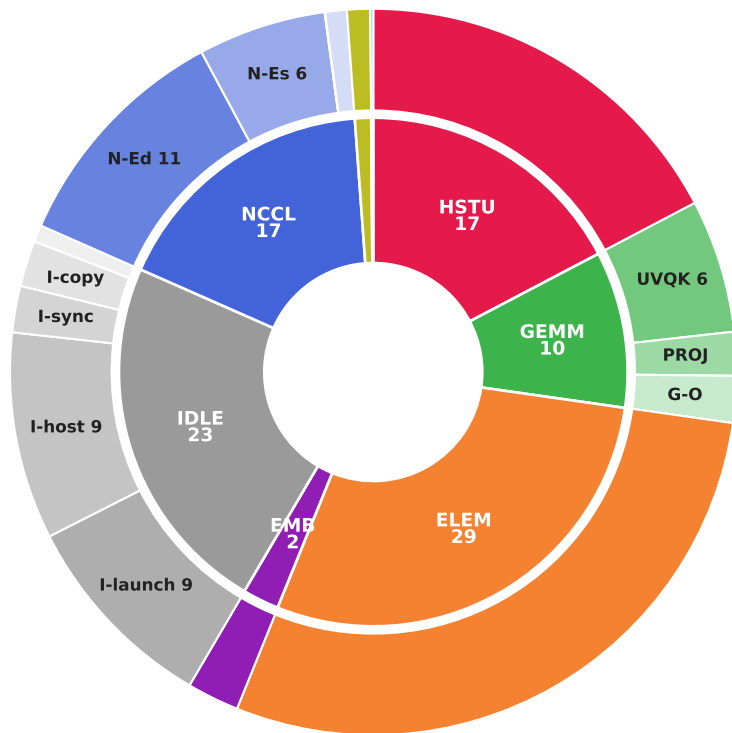
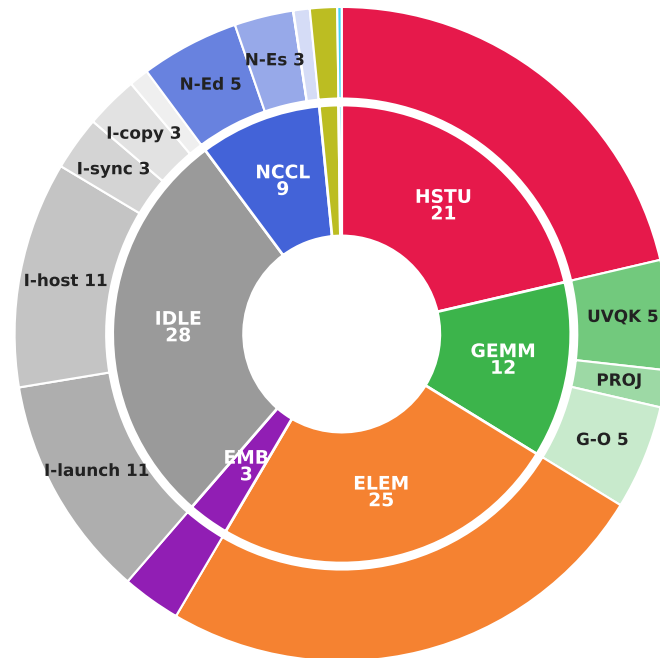


§8a v26.05 → v26.06 image A/B — GPU-time sunburst, matched 1B/4096/lognormal (8-GPU, 80-step aggregate)
 inner ring = grouping · outer ring = leaves · % of step; DISK AREA $\propto$ step time (v26.06 = 0.81× area) → equal area = equal absolute ms
 step −18.9% · idle area ~flat (launch+host unchanged) · exposed NCCL ~halves · elementwise shrinks

v26.05 (median 94.0 ms/step · MFU ~16%)



v26.06 (median 77.5 ms/step · MFU ~19%)



■ HSTU = hstu fwd/bwd
 ■ G-O = gemm / others
 ■ N-Ed = nccl DENSE/all-reduce (exposed)
 ■ I-launch = idle: kernel-launch (CPU dispatch)
 ■ I-sync = idle: sync wait
 ■ OTH = others
■ UVQK = gemm / uvqk
■ ELEM = elementwise
■ N-Es = nccl SPARSE/all-to-all (exposed)
■ I-host = idle: host/python gap
■ I-copy = idle: H↔D copy
■ OVL = overlapped
■ PROJ = gemm / projection
■ EMB = embedding op
■ N-Os = nccl sparse (overlap)